TEAM – 5

Prediction of Silica Concentration in Iron Ore Mining

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Abstract: The Extraction of iron ore is a essential and crucial process in the modern, industrial world. The quality of iron ore is very essential and of importance in terms of economic viability in order to assess purity, quality of final product and affects. Using Linear Regression, Random Forest, K-Nearest Neighbours, Support Vector Regressor and Polynomial Regression, Decision Tree in order to predict silica concentration in iron ore mining, to compare and find out the best to forecast silica levels.

Keywords: Extraction, Iron ore, Silica concentration, Regression models, Mining prediction

1. INTRODUCTION

Iron ore mining is essential for the metallurgical industry, supplying the raw materials for steel production. However as impurities will be present particularly the presence of silica posses a challenge in order the gain a specific product specification. The smelting process and quality of steel overall are impacted due to the presence of silica. Hence accurate predictions are of utmost importance for mining and processing.

Traditional methods often have very time consuming laboratory analysis and testing making both lot of time spent and hard decision making very challenging. Machine Learning algorithms by comparison provide a more quicker and faster alternative making the process a lot quicker. Utilizing Linear Regression, Random Forest, kNN, SVM, and Polynomial Regression to predict the silica concentration and to compare the performance of each in order to find the most efficient and effective model.

Linear Regression provides a simple and interpretable baseline, Random forest uses ensemble learning in order to capture the various complex relationships of the data, SVM is excellent in handling non-linear relationships and Polynomial Regression catches the higher-order interactions. By comparing these models strengths and weakness of each will be proved and highlighted in regards to predictions of silica concentration in iron ore and the valuable insights and information that can be provided by each of these models as well as they're working.

2. LITERATURE REVIEW

2.1 Background

This study focuses on impact heating processes and mining operations, where using multi-target regression techniques to predict quality is becoming more and more important. Mineral concentrations including silica and iron are among the secondary factors that are evaluated in the analysis. The study is noteworthy for its investigation into the simultaneous prediction of iron and silica concentrations, which emphasizes the intrinsic connection between these variables. This is accomplished by thoroughly comparing a variety of regressors, including Decision Tree, AdaBoost, Random Forest, and k-Nearest Neighbors algorithms, to identify the best model. These models' performance is evaluated using evaluation measures, most notably the coefficient of determination (R2). Because the discovery goes beyond conventional methods that anticipate these concentrates independently, it makes a substantial contribution.

2.1 Background

Apart from mining, the research explores the complex dynamics of blast furnace operations, with an emphasis on aberrations concerning furnace mechanisms to maximize the output of hot metal. Furnace temperature feature data is made less dimensional by applying novel techniques like least redundant mutual information feature selection and maximum correlation. Balanced Proximity Gray Correlation Analysis is used in the study to find correlations between molten iron silicon content and furnace temperature parameters at various time delays. A parameter-based network connecting several control parameters to numerous state parameters is established by integrating furnace temperature characterization tensors and then parameterizing the system using the GRA-FCM model. The goal of the all-encompassing method is to use the GWO-SVR model to dynamically anticipate the silicon concentration of molten iron, allowing control parameters to be adjusted in real-time for accurate furnace temperature control. The study investigates the use of artificial neural networks (ANN) for predicting silicon content, with encouraging findings in terms of mathematical correlation and mean square error. The significance of silicon management in foundry production is acknowledged.

2.2 Related works

This writing study bridges the gaps between metallurgy and mining operations by highlighting the interdisciplinary aspect of the inquiry. It emphasizes how crucial dynamic control models and sophisticated regression techniques are to raising the effectiveness and caliber of metal processing and blast furnace operations. Optimizing blast furnace operations and ore treatment in the mining industry requires the deployment of multi-objective regression techniques and advanced predictive modeling. The ability to forecast iron and silica concentrations simultaneously is a novel addition that acknowledges the intricate relationship between these two variables. In the context of blast furnace dynamics, the study emphasizes the value of novel approaches like feature selection and correlation analysis in lowering dimensionality and creating significant correlations. The overall goals of the research are to enhance ore quality and advance knowledge of the dynamic hot metal creation in impact heater operations.

2.3 Motivation (Optional)

This comprehensive literature review highlights the application of advanced regression techniques and dynamic control strategies, including multi-objective regression and neural networks, to predict silica and iron concentrations in combined ore processing, blast furnace operation optimization and overall efficiency improvement in the mining industry.

3. PROPOSED METHODOLOGY

3.1 Data Preprocessing

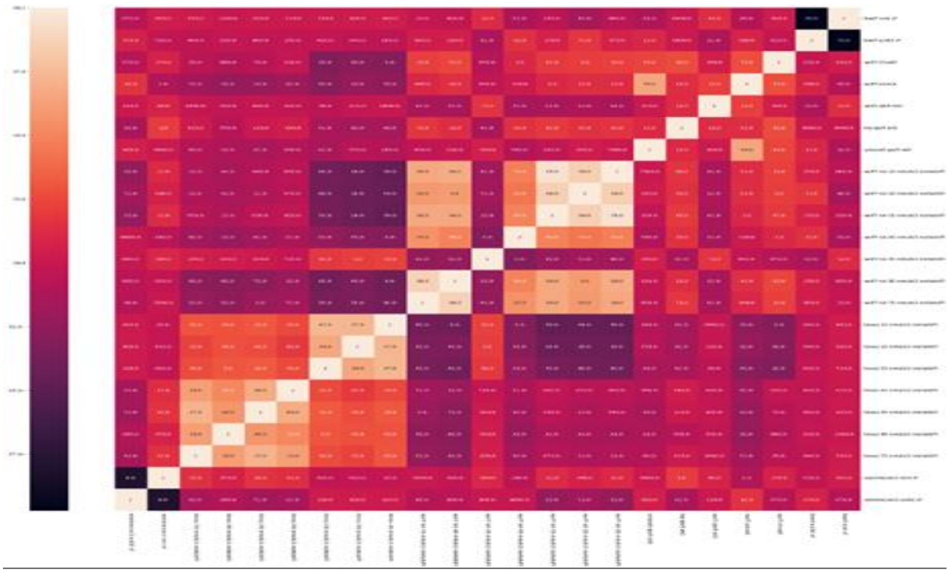

Fig.1 Heat Map of the Dataset Features.

Impurities in silica involve alumina and iron oxides hurt the overall quality. These impurities affect the separation process, affecting the efficiency of silica removal.

Overfitting in froth floatation data happens when a model learns the specific patterns or noise of data that do not generalize effectively. This leads to poor predictive analysation and affects complexity control of the model in order to not only read the data properly but also ensure proper predictions.

Noisy data in starch and anima flow arises from inaccuracies and fluctuations in data. Noise hurts the predictive nature and reliability of these parameters, leading to suboptimal performances and hurts critical measurements.

3.2 Methodologies or Algorithms

3.2.1 LINEAR REGRESSION

Linear regression is a statistical model that establishes a relationship between dependent and independent variables by fitting linear equations to the data of silica in iron ore. The goal is to minimize the difference in predicted and actual values and establish a relationship between the two. Linear regression in Silica detection can be used with various input features like Ore density, air flow rate, pH levels. Using these features trained model can then predict the silica amount based on new inputs. Real time predictive capacity and iron ore processing is established and can be used to optimize the removal of silica impurities in froth floatation.Linear Regression provides a baseline model that provided straightforward and interpretable information particularly used for initial analysis. The effectiveness depends on the linearity of the relationship of the data.

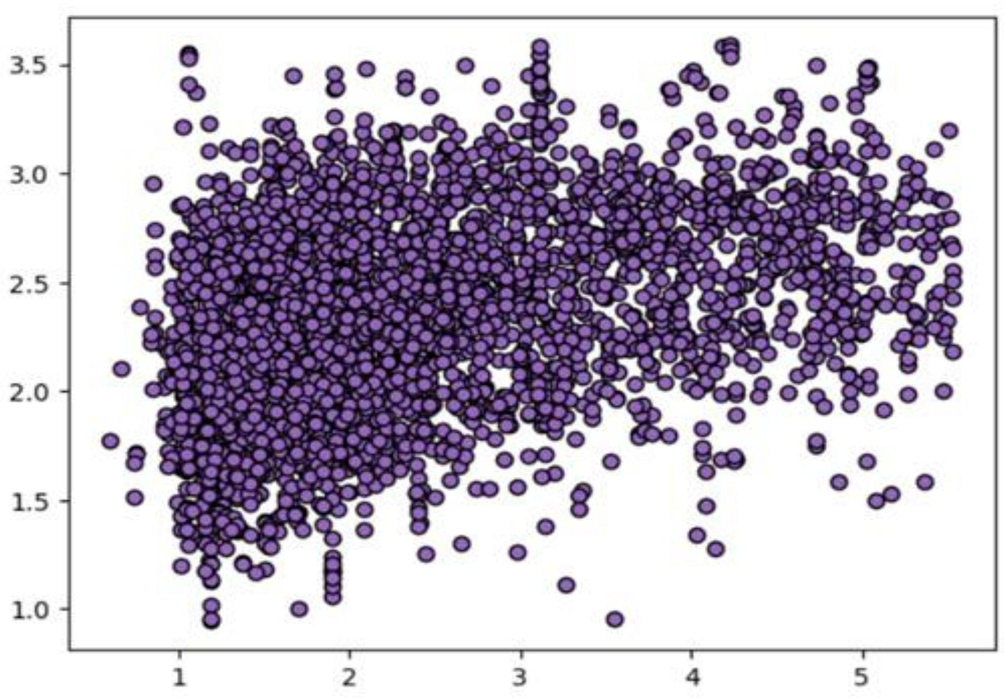

Fig.2 Scatter Plot of Predicted Data vs Labeled Data using Linear Regression.

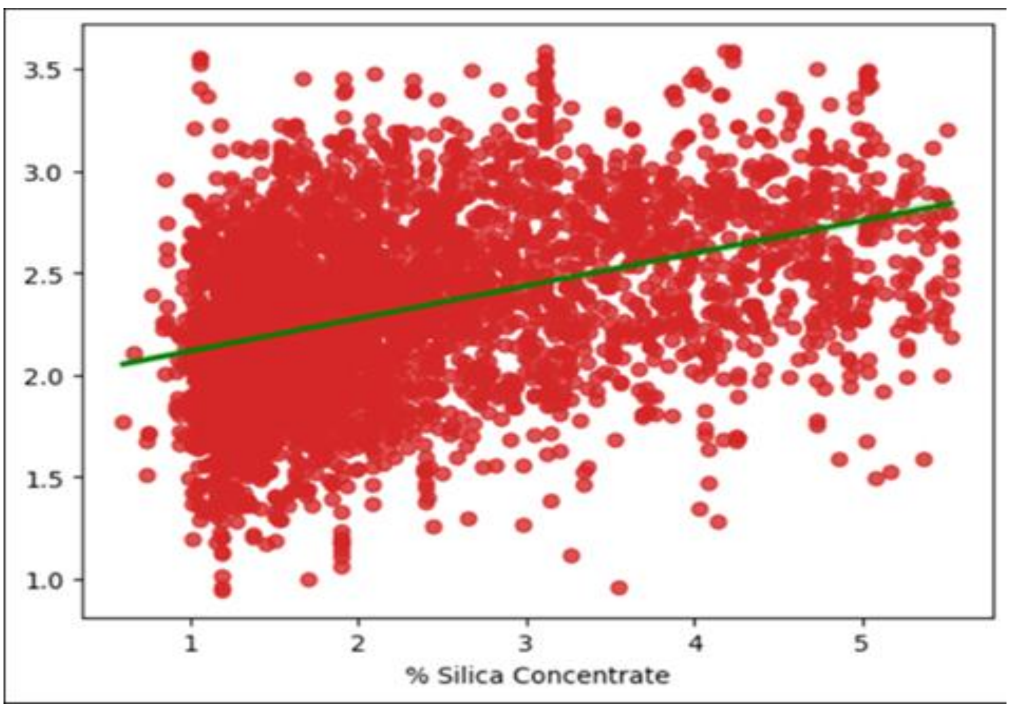

Fig.3 Regression Plot of Predicted Data vs Labeled Data using Linear Regression.

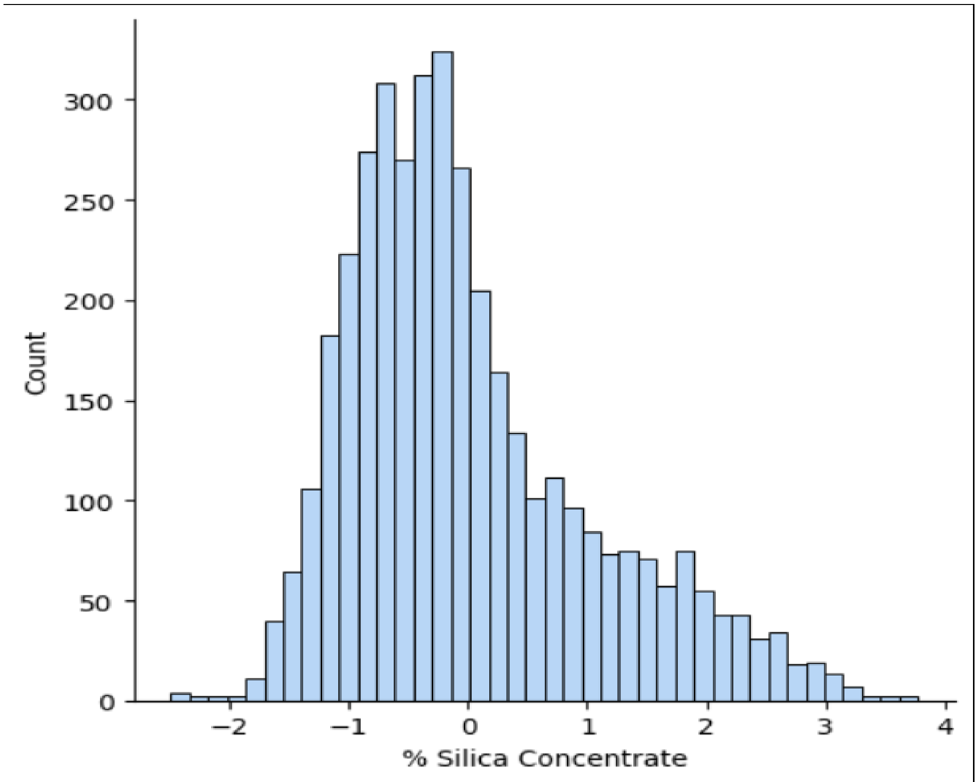

Fig.4 Histogram Plot of Predicted Data vs Labeled Data using Linear Regression.

3.2.2 KNN REGRESSOR

k-Nearest Neighbours Regressor is a supervised machine learning algorithm used for regression tasks. Regression is to predict continuous numerical values and discrete class values. K training instances (data points) that are the closest to one another and to the new data point in the feature space. The average of the new data point or weighted average of the outputs from the k nearest neighbours are taken to predict the output.

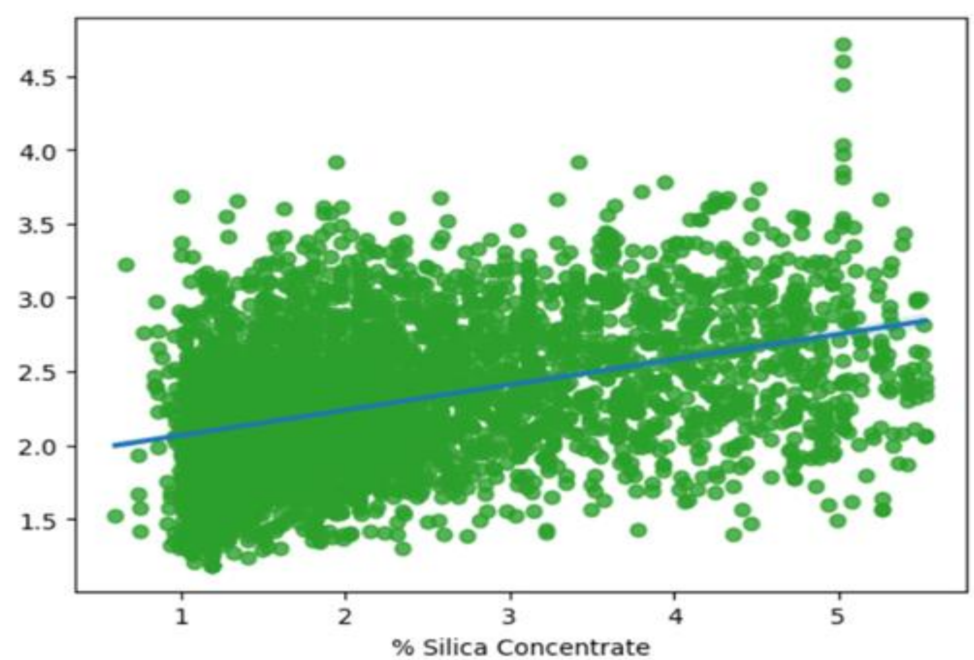

Fig.5 Regression Plot of Predicted Data vs Labeled Data using KNN Regression.

Knn regression can capture various complex relationships in data. Here it captures the relationship between the input features and silica concentration and if its non-linear. Knn is however sensitive in regards to distance metric and value of k. Knn uses parameter tuning which is selecting the right and optimal values for the hyperparameter and cross validation which checks the performance and generalization ability.

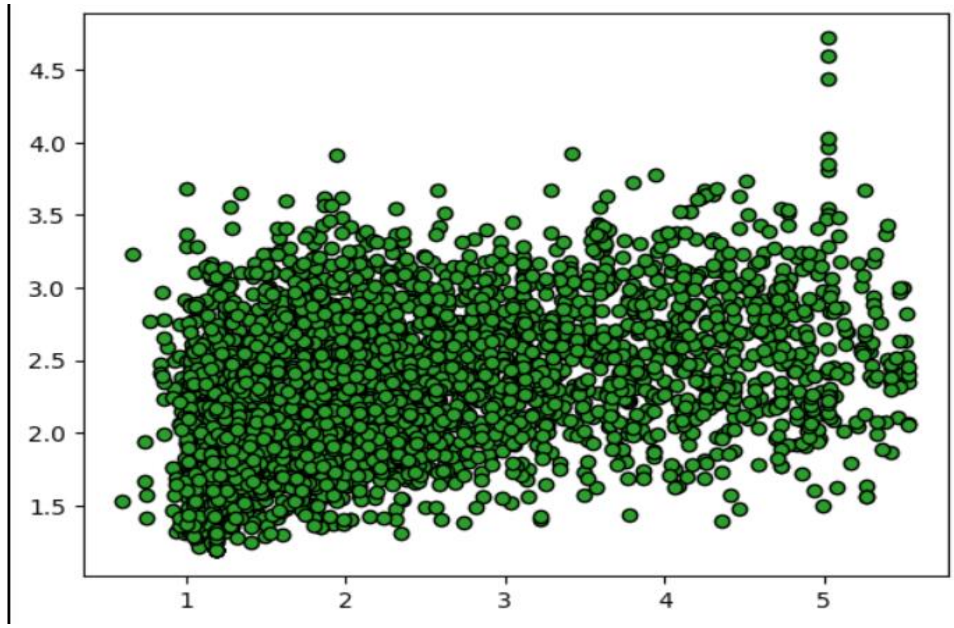

Fig.6 Scatter Plot of Predicted Data vs Labeled Data using KNN Regression.

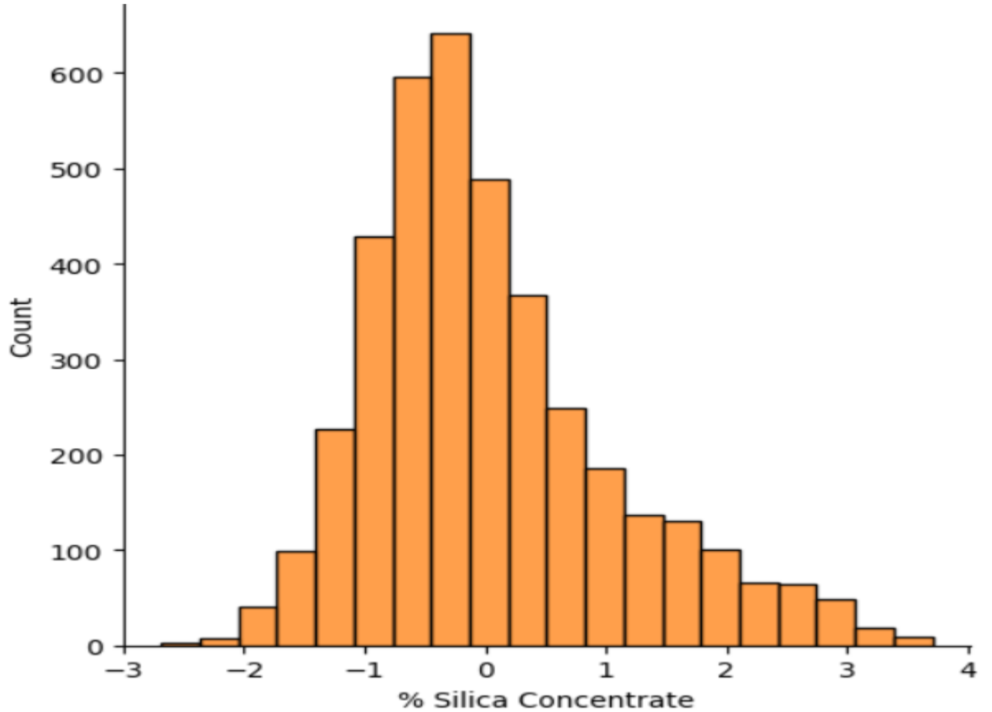

Fig.7 Histogram Plot of Predicted Data vs Labeled Data using KNN Regression.

3.2.3 RANDOM FOREST REGRESSOR

Random Forrest Regressor is a machine learning algorithm that uses multiple decision trees to improve the overall accuracy and generalization. Using a multiple amount of decision trees the average of regression tasks of each tree is taken into account and the average is derived from it for the output.

Random Forrest Regressor captures the complex relationship between the many input features ex air flow rates and Ph values, and their interactions. Handling of non linearities along with analysis of features makes it able to provide effective in as it analyses and makes accurate predictions of silica levels in froth floatation process in order leading to better decision making overall.

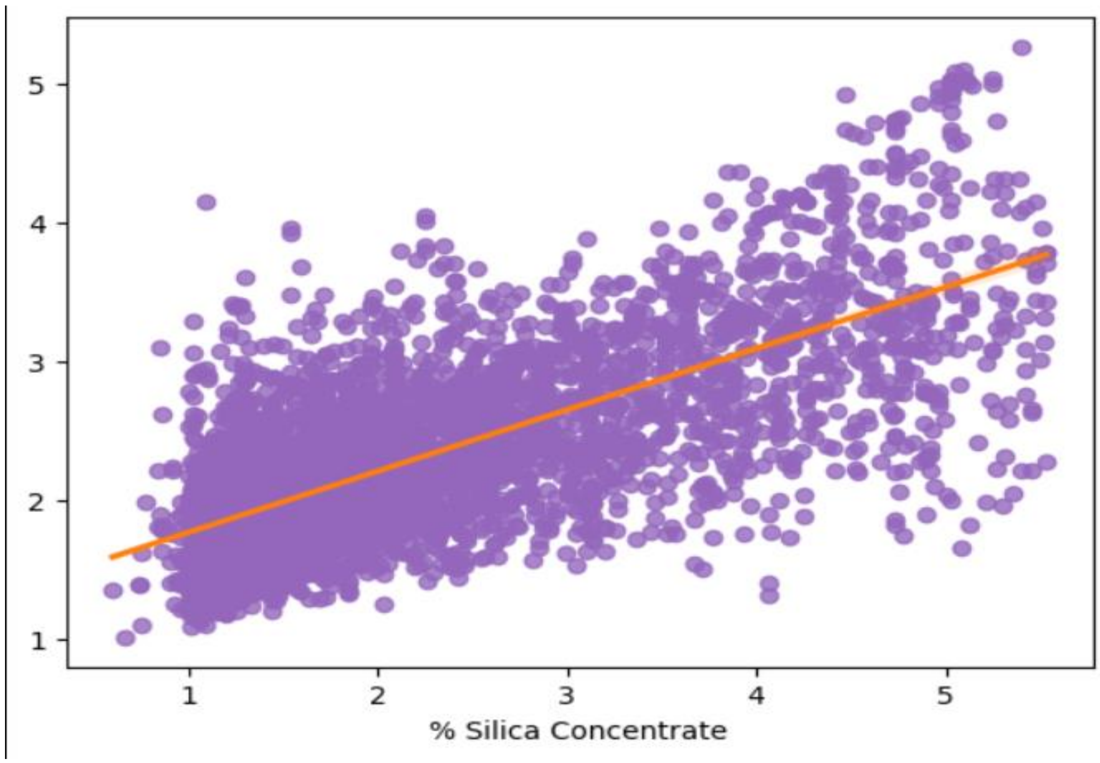

Fig.8 Regression Plot of Predicted Data vs Labeled Data using Random Forest Regression...

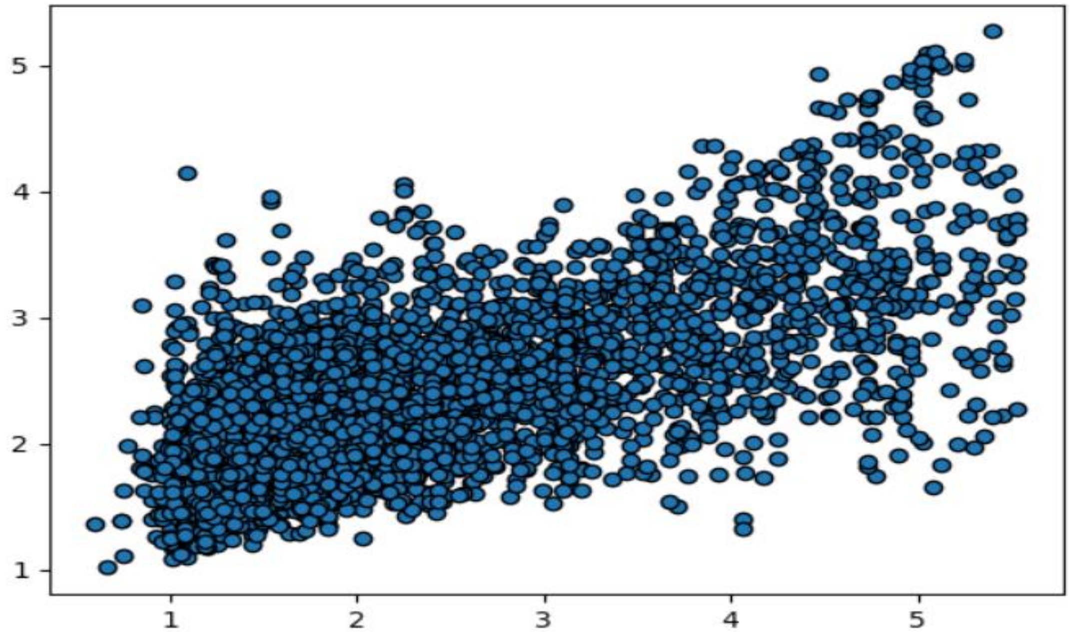

Fig.9 Scatter Plot of Predicted Data vs Labeled Data using Random Forest Regression..

Using a feature importance in addition of optimizing the froth floatation process, Random Forrest Regressor has a keen and good understanding of the data and is able to predict values of silica content in iron ore effectively. Often gives insights into feature importance and with the help of the multiple decision trees during training the data, each tree helps with final prediction and give very reliable and accurate prediction reducing influence and any mitigated risks overall there by giving accurate and enhance data. Critical parameters are highlighted and adjusted during ore benefication.

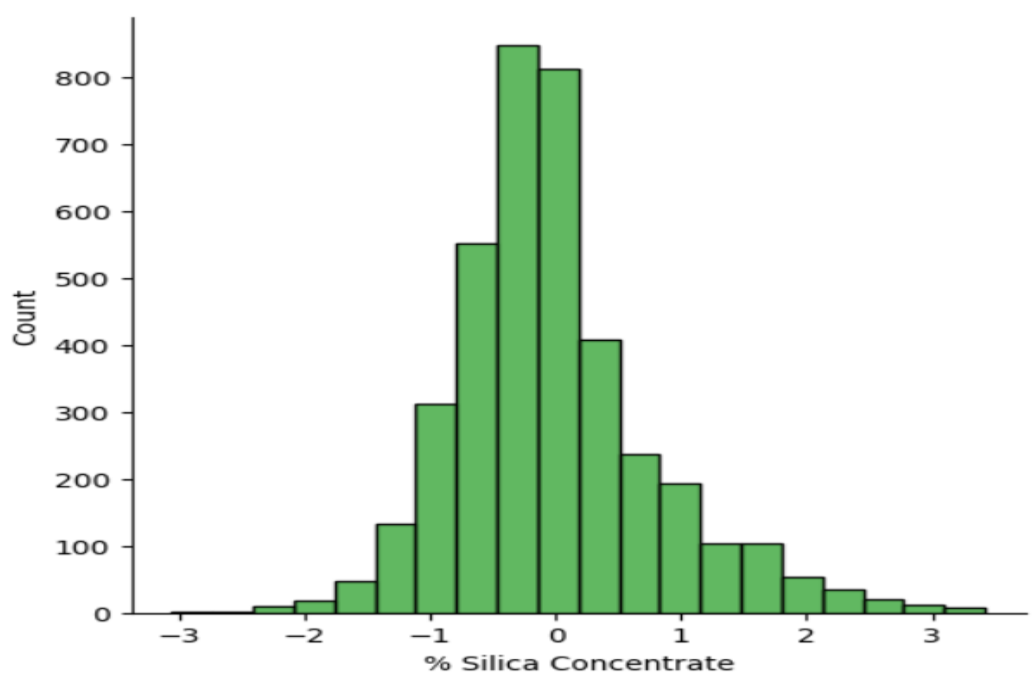

Fig.10 Histogram Plot of Predicted Data vs Labeled Data using Random Forest Regression..

3.2.4 DECISION TREE REGRESSOR\

Decision Tree is a tree model representing decisions and their possible consequences. The dataset is broken down into many smaller portions or subsets with each having specific directions. Each feature is represented by a node of the tree and each respective leaf the predicted outcome.

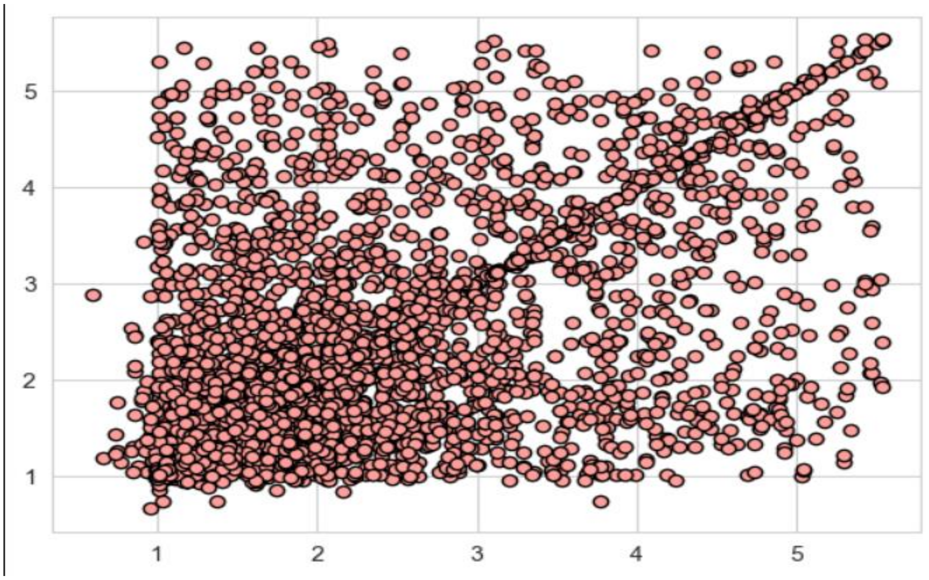

Fig.11 Scatter Plot of Predicted Data vs Labeled Data using Decision Tree Regression..

Decision tree provides a clear model for the understanding of the relationships in the data. Between input features and silica levels. Decision Tree are excellent at identifying critical factors in this case ex. Ore composition. At each node the decision-making process is represented at each node. This helps in understanding key parameters that are of importance that help in prediction of silica concentration in iron ore. The model helps in giving a deep and insightful look with each node of the relationship that may be challenging for other linear models to achieve. After being trained Decision Trees can be used to predict better and frequently. Branches of the tree provide current values and adjustments to the parameters can be made.

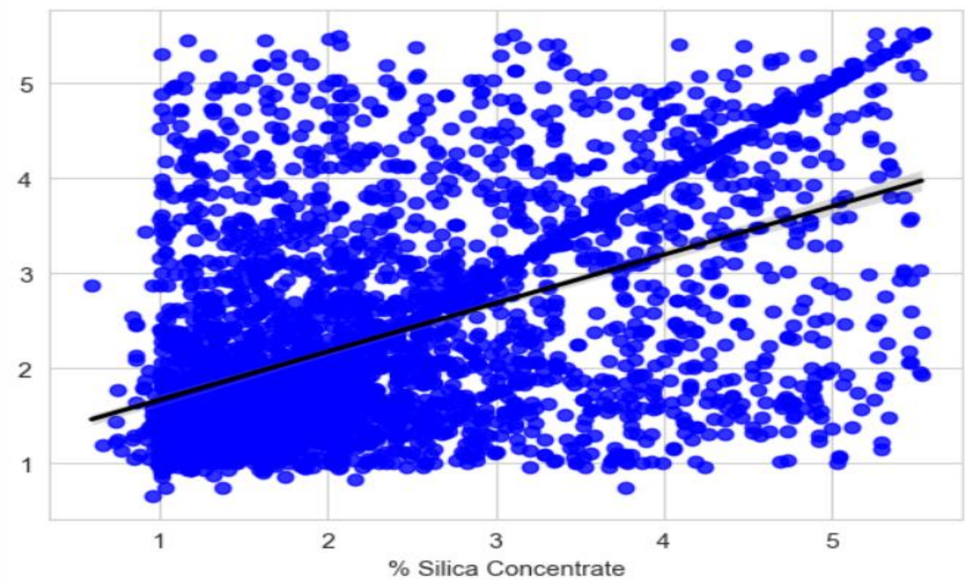

Fig.12 Regression Plot of Predicted Data vs Labeled Data using Decision Tree Regression...

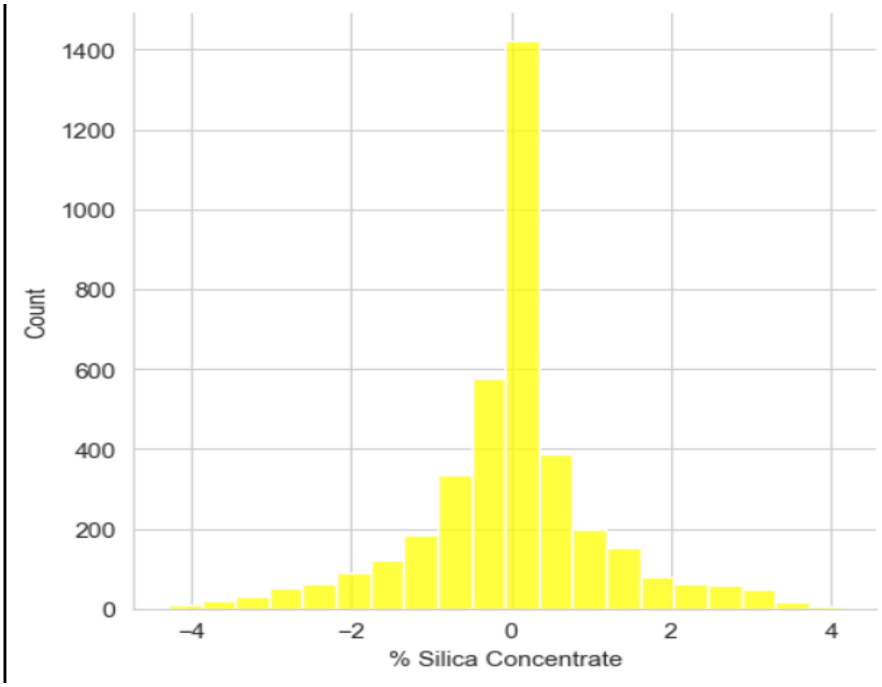

Fig.13 Histogram Plot of Predicted Data vs Labeled Data using Decision Tree Regression..

3.2.5 SUPPORT VECTOR REGRESSOR

A support vector regressor is generally used for regression tasks and is utilised to predict continuous numerical values. The hyperplane is found between the best way to represent the relationship between input features and target variable. The errors are to be minimized to generate greater accuracy and help out with greater prediction and usage. Error must be minimized between the actual and predicted values to maintain a level of balance and maintaining the margin of tolerance between the two values respectively.

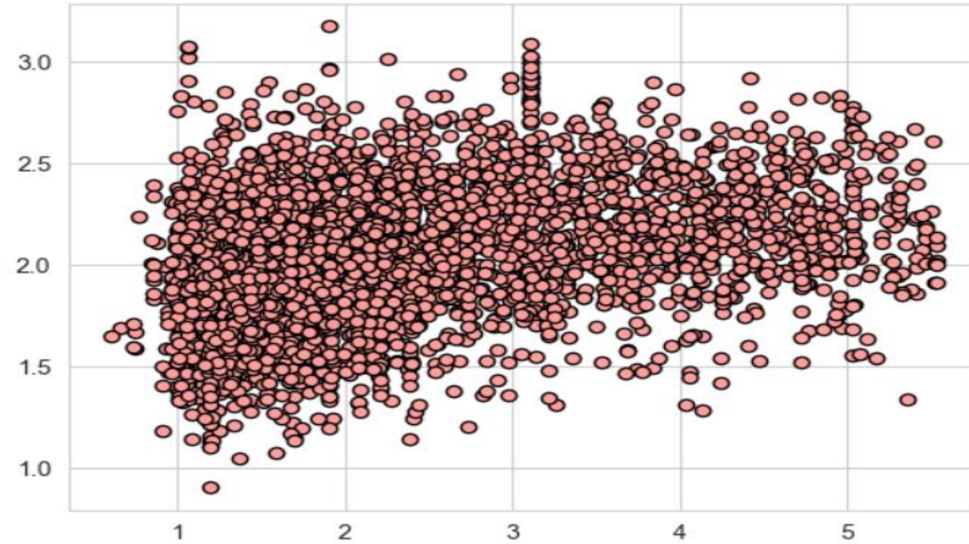

Fig.14 Scatter Plot of Predicted Data vs Labeled Data using Support Vector Regression..

For silica concentration Support Vector Regression proves very useful as it handles very intricate data and well as it deals with non-linear relationships. SVR captures the relationships well and excels in understanding complex patterns and understands various input parameters.

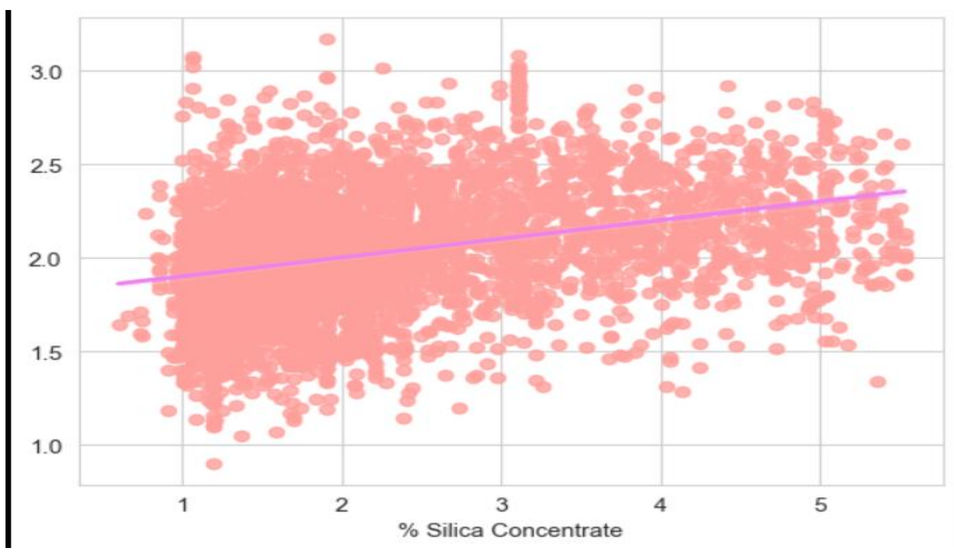

Fig.15 Regression Plot of Predicted Data vs Labeled Data using Support Vector Regression...

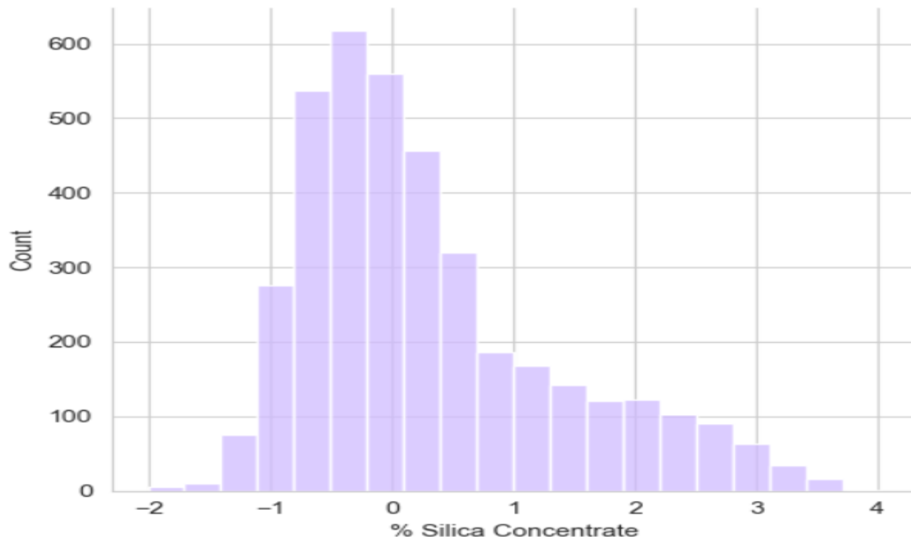

Fig.16 Histogram Plot of Predicted Data vs Labeled Data using Support Vector Regression...

SVR works very effectively to ensure a comprehensive analysis of multiple influencing factor of silica prediction in iron ore extraction using high dimensional spaces. SVR can adapt to a variety of data variations which in turn leads to better decision making with regard to removal of silica in iron ore.

3.2.6 POLYNOMIAL REGRESSION

Polynomial regression uses the relationship between independent and dependent variable which is then modelled as nth degree polynomial. Polynomial regression allows many complex and non linear patterns unlike linear relationships with the help of polynomial terms. The model takes on the equation of

y=f(x)=β0+β1x+β2x2+β3x3+⋯ +βdxd+ + β d x d + ϵ .

where y serves as the dependent variable and x serves as the dependent variable.

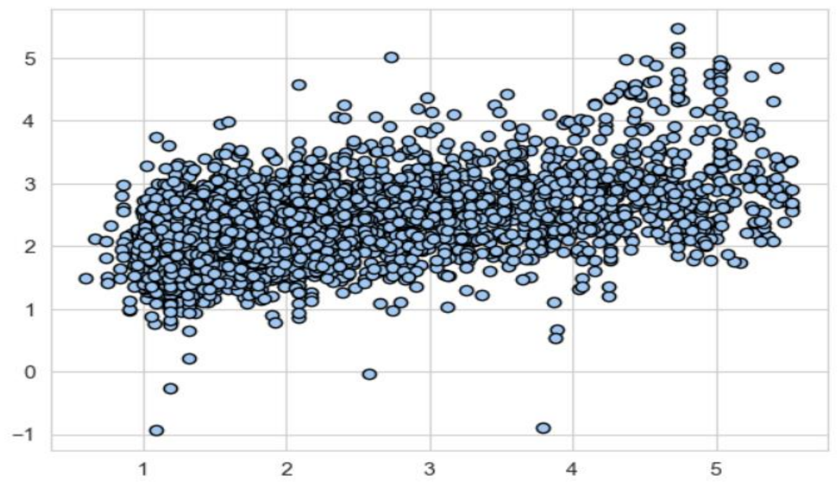

Fig.17 Scatter Plot of Predicted Data vs Labeled Data using Polynomial Regression...

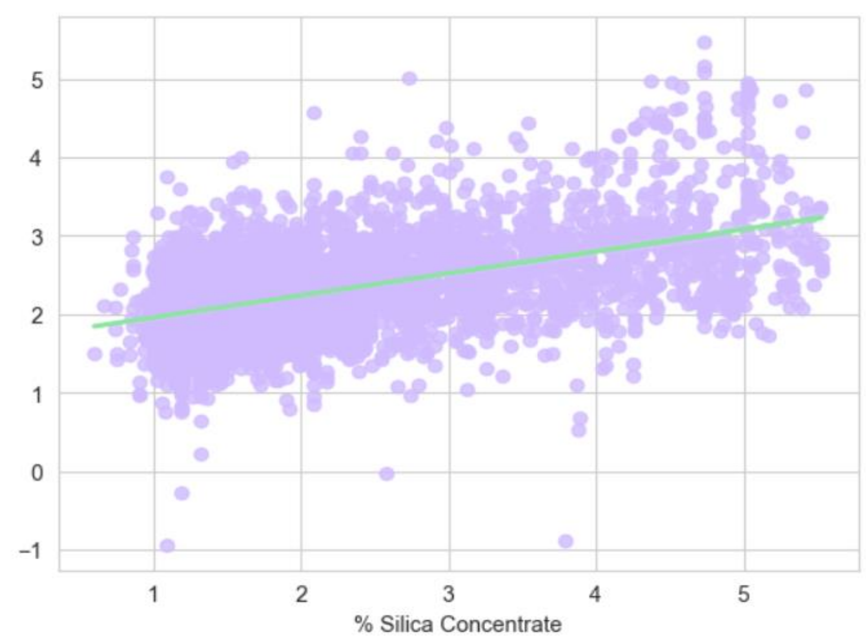

Fig.18 Regression Plot of Predicted Data vs Labeled Data using Polynomial Regression....

Polynomial Regression has non linear relationships between input features and the silica value in the ore.Polynomial Regression possess higher order polynomial terms, by using these terms the complexities of the data are captured in a more effective manner, it can also capture intricate and curved patterns often which are often present. Thereby providing a more effective predictive model that is better at overall representation.

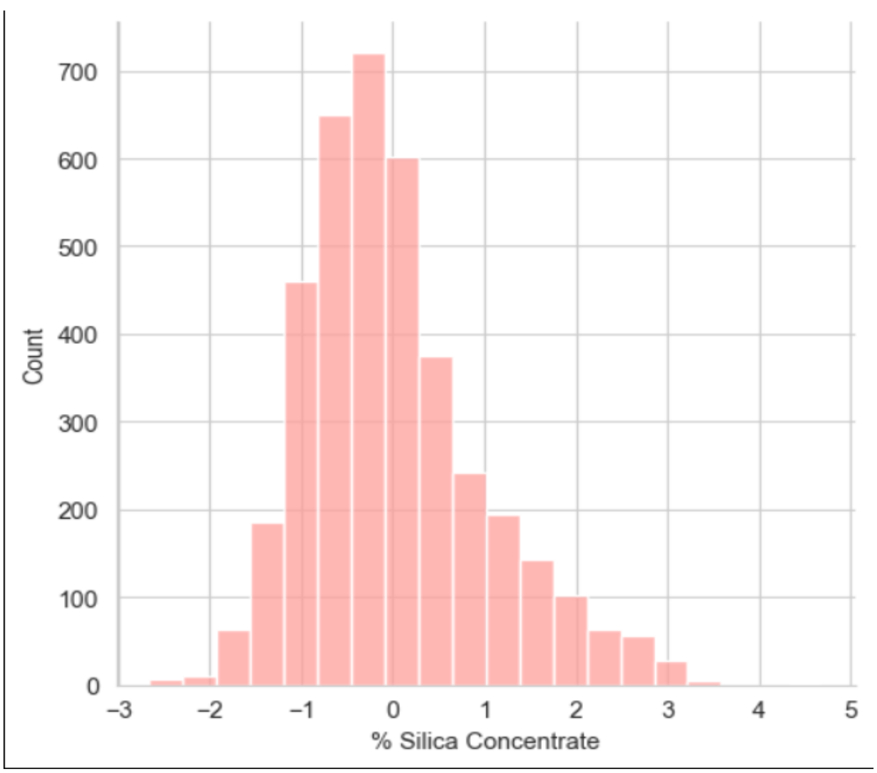

Fig.19 Histogram Plot of Predicted Data vs Labeled Data using Polynomial Regression...

3.3 Proposed Architecture

Example for work flow of proposed architecture

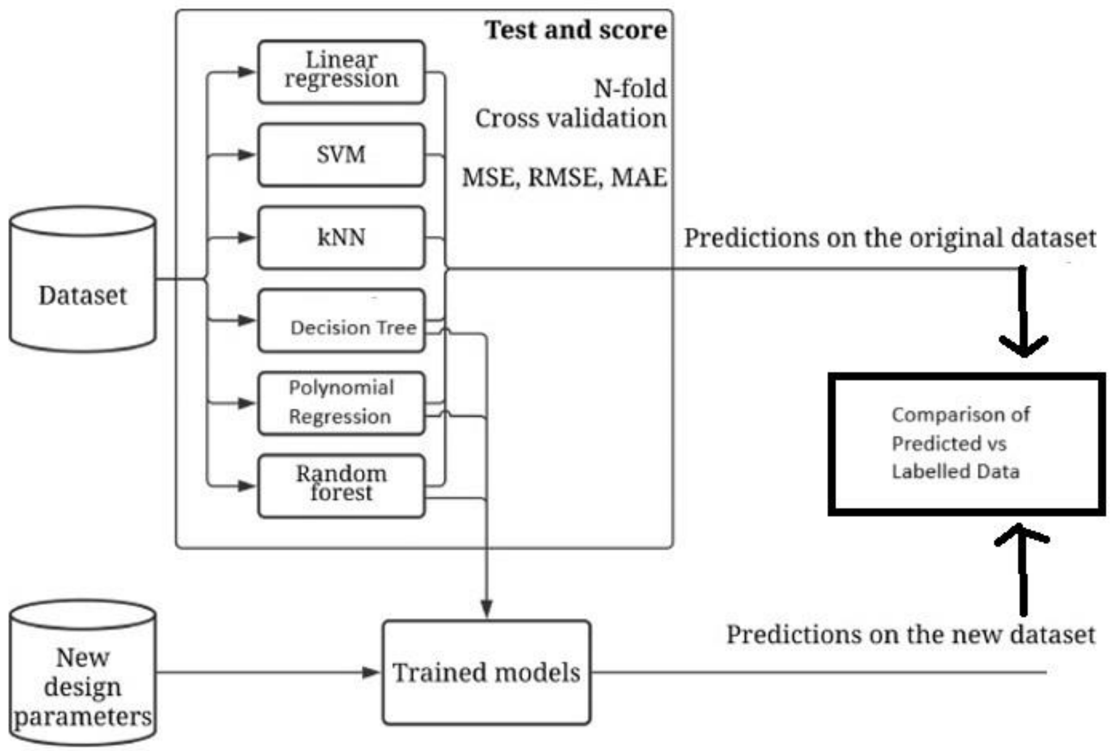

Fig.20 Froth Flotation Process – for Silica Concentration – ML Architecture.

Example for real time environment

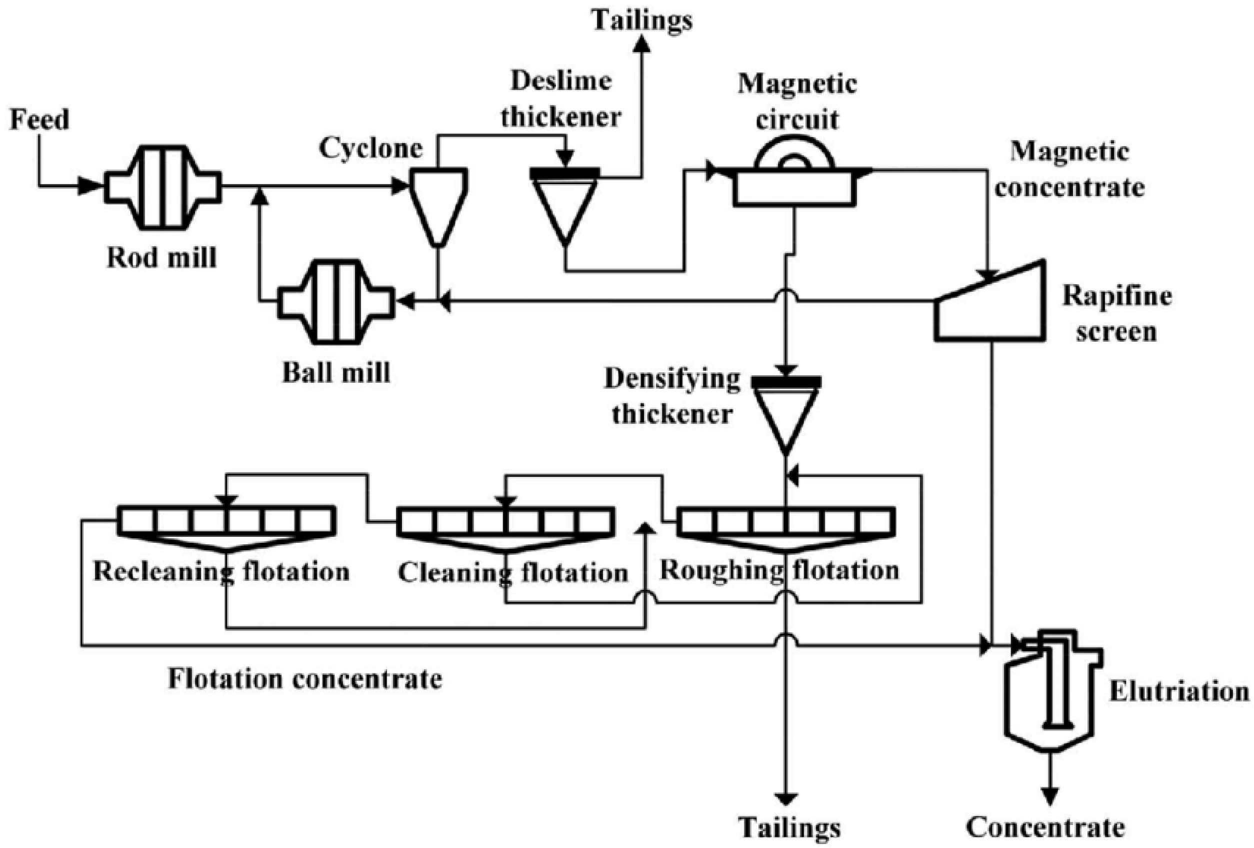

Fig.21 Froth Flotation Process – for Silica Concentration.

Example for real time environment

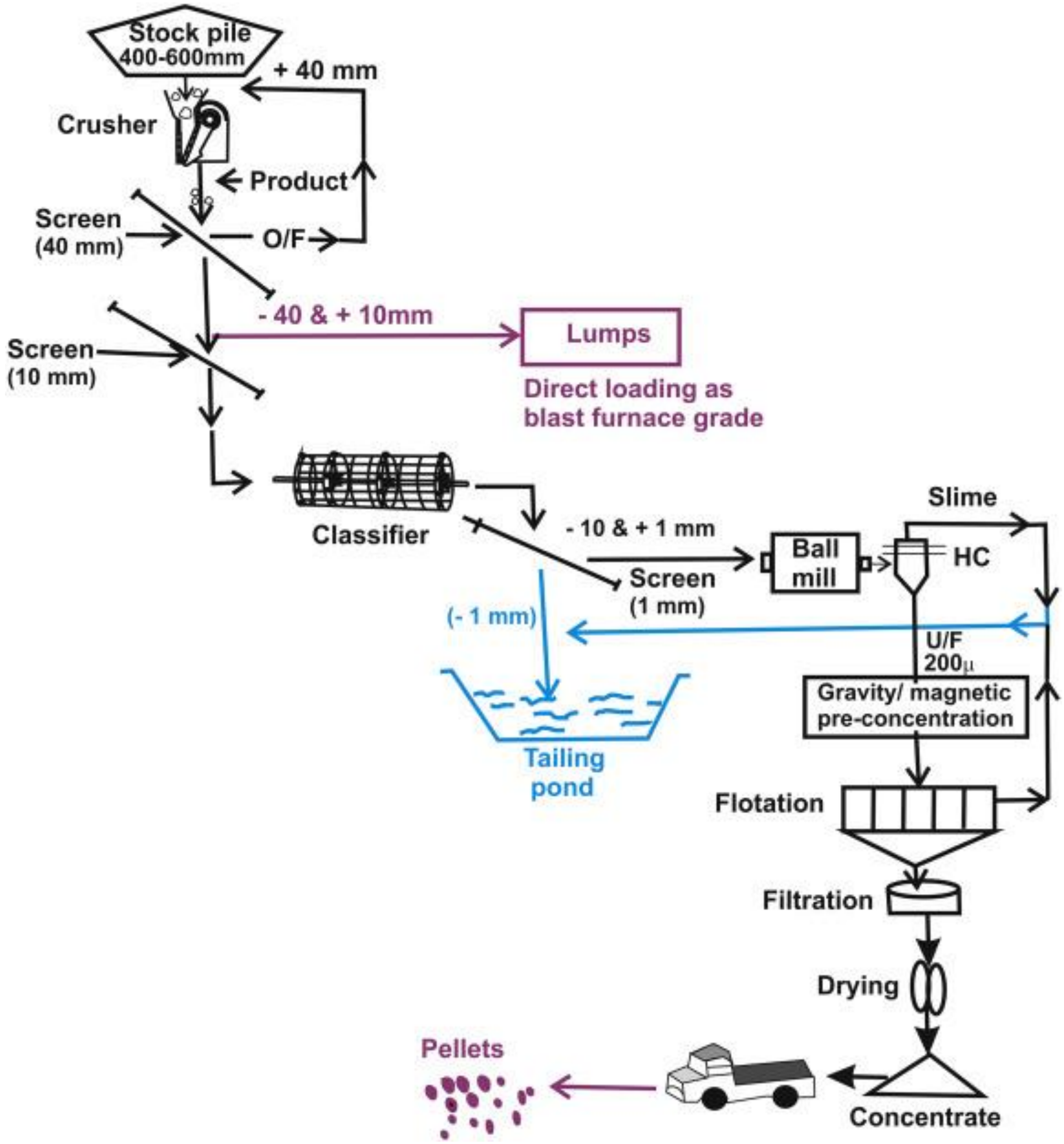

Fig.22 Froth Flotation Process – for Silica Concentration.

4. EXPERIMENTAL ANALYSIS AND RESULTS DISCUSSION

4.1 Experimental setup and Dataset

Features used here are Floatation column, Ore pulp density, Air column, silica feed, Iron feed, Ore Pulp ph. and Anima Flow. The floatation column represents the liquid levels at different sections of the floatation column. The Ore pulp density shows the concentration of ore particles in pulp and how it influences the froths stability and separation efficiency. Silica feed and iron feed constitute the initial composition of the ore and helps to predict froth floatation. Air flow refers to floatation columns at different air flow rates. Ore Pulp ph refers to the acidity or alkalinity of the pulp and gives various insights. Anima flow shows the movement of air bubble and their impact in the froth from both stability and effectiveness.

Ore pulp pH is of utmost importance due to the chemical alkalinity or acidity of the ore, Silica and Iron feed % provide us with useful contents of the pulp hence these 2 features are of utmost importance and use.

Impurities in silica involve alumina and iron oxides hurt the overall quality. These impurities affect the separation process, affecting the efficiency of silica removal.

Overfitting in froth floatation data happens when a model learns the specific patterns or noise of data that do not generalize effectively. This leads to poor predictive analysation and affects complexity control of the model in order to not only read the data properly but also ensure proper predictions.

Noisy data in starch and anima flow arises from inaccuracies and fluctuations in data. Noise hurts the predictive nature and reliability of these parameters, leading to suboptimal performances and hurts critical measurements.

4.2 Evaluation parameters and formulations

Mean Absolute Error (MAE), Mean Square Error (MSE), and Root Mean Square Error (RMSE) are the assessment parameters that were applied to the data that was supplied. These measures are frequently used to evaluate regression models' performance. Here's a quick rundown of each

The average absolute difference between the expected and actual values is measured by the Mean Absolute Error (MAE). Without taking into account the direction of the mistakes, it provides a general indication of their average size.

Mean Square Error (MSE): It computes the mean of the squared discrepancies between the values that were expected and those that were observed. MSE is susceptible to outliers since it assigns greater weight to bigger mistakes.

Root Mean Square Error (RMSE): This statistic, which measures the average magnitude of errors in the same units as the predicted variable, is the square root of the mean square error (MSE). When mistakes are anticipated to be both positive and negative, RMSE is frequently employed.

The Random Forest Regressor (RFG), which shows the lowest values for each of the three evaluation parameters (MAE, MSE, and RMSE), is determined to be the model with the greatest performance based on the data presented. The justification focuses on the advantages of the Random Forest Regressor, including its capacity to identify intricate linkages, effectively manage non-linear patterns, and profit from the ensemble forecasts produced by several decision trees.

4.3 Comparison methods

To estimate the silica content in iron ore, a number of regression models were used in the analysis. The Mean Absolute Error (MAE), Mean Square Error (MSE), and Root Mean Square Error (RMSE) were used to evaluate each model. The models that are taken into consideration are Decision Tree, Support Vector Regressor (SVR), k-Nearest Neighbors (Knn), Polynomial Regression, Random Forest Regressor (RFG), and Linear Regression. With the lowest MAE of 0.608, MSE of 0.664, and RMSE of 0.815 among these models, the Random Forest Regressor stands out as having the best predictive abilities. The Random Forest Regressor's ensemble nature, which combines predictions from several decision trees for improved generalization, helps it perform exceptionally well at capturing intricate relationships and non-linear patterns.

4.4 Results and discussion

As noted from the below data Random Forrest Regressor shows the lowest Mean Absolute Error with 0.608 and lowerst Mean Square Error with 0.664 and lowest Root Mean Square Error with 0.815, hence proving Random Forrest Regressor higher capability . Random Forrest Regressor captures complex relationships and non linear patterns very efficienty and effectively. As Random Forrest Regressor possess an ensemble nature predictions from multiple decision trees allows further generalisation and better overall in regards for prediction of silica concentration in iron ore overall .

Name	MAE	MSE	RMSE
Linear	0.810	1.065	1.032
RFG	0.608	0.664	0.815
Knn	0.794	1.073	1.036
SVR	0.804	1.215	1.102
Polynomial	0.748	0.940	0.969
Decision Tree	0.721	1.253	1.119

MAE – Mean Absolute Error

MSE – Mean Square Error

RMSE – Root Mean Square Error

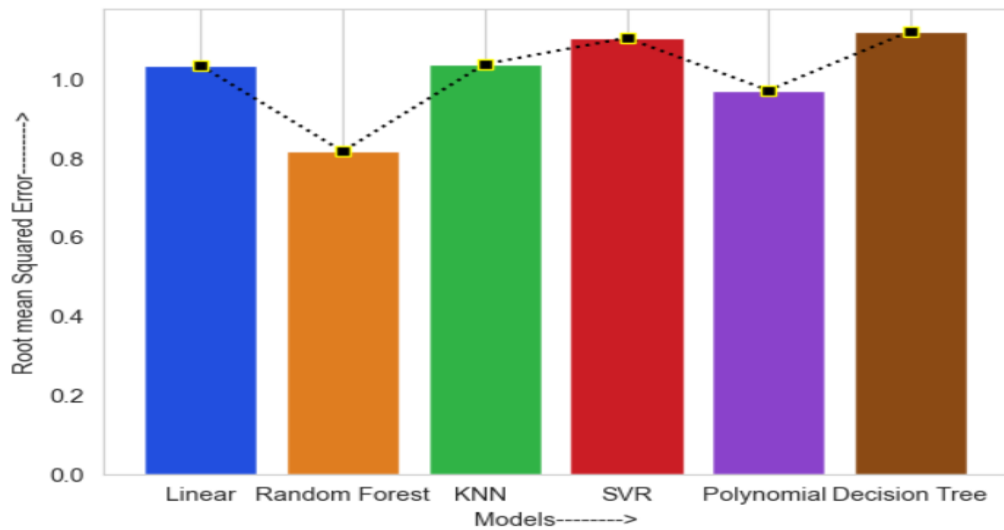

Fig.20 Root Mean Squared Errors of the algorithms used for Regression Analysis.

5. CONCLUSION

Hence using Linear Regression, Random Forest, K-Nearest Neighbours, Support Vector Regressor and Polynomial Regression, Decision Tree for Prediction of Silica Concentration in Iron Ore Mining. From the tests run and outcomes reached Random Forrest Regressor proved most effective and efficient with lowest Mean Absolute Error, Mean Square Error, Root Mean Square Error and a R2 score of 0.159. Hence proving Random Forrest as the most accurate and optimal model from the above.

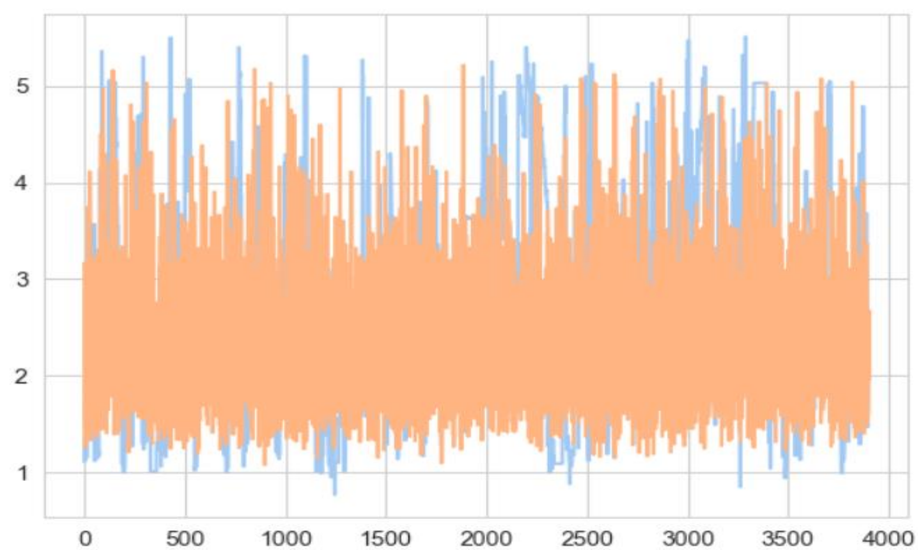

Fig.20 Random Forrest Regressor vs Labelled values.

Due to the higher Error scores of the other models and sacrifices in important departments and as Random Forrest could provide an overall higher accuracy range and faster predications along with using root and nodes provide more scenarios with the real world thereby enhancing its overall capability and efficiency using more accurate silica amounts and better overall performance in the end.

REFERENCES

[1] Cui, Z., Yang, A., Wang, L. and Han, Y., 2022. Dynamic Prediction Model of Silicon Content in Molten Iron Based on Comprehensive Characterization of Furnace Temperature. Metals, 12(9), p.1403.

[2] Nagarjuna, P. and Ananthan, T., 2022, October. Machine Learning Based Estimation of Silicon Concentration in Froth Flotation Process. In 2022 IEEE 3rd Global Conference for Advancement in Technology (GCAT) (pp. 1-6). IEEE.

[3] Dogan, A., Birant, D. and Kut, A., 2019, September. Multi-target regression for quality prediction in a mining process. In 2019 4th international conference on computer science and engineering (UBMK) (pp. 639-644). IEEE.

[4] Cardoso, W., Di Felice, R., Dos Santos, B.N., Schitine, A.N., Machado, T.A.P., Galdino, A.G.D. and Dixini, P.V.M., 2022. Modeling of artificial neural networks for silicon prediction in the cast iron production process. IAES international journal of artificial intelligence, 11(2), p.530.

[5] Zhou, S., Yuan, Z., Cheng, Q., Weindorf, D.C., Zhang, Z., Yang, J., Zhang, X., Chen, G. and Xie, S., 2020. Quantitative analysis of iron and silicon concentrations in iron ore concentrate using portable X-ray fluorescence (XRF). Applied spectroscopy, 74(1), pp.55-62.

[6] Fang, Y., Jiang, Z., Pan, D., Gui, W. and Chen, Z., 2020. Soft sensors based on adaptive stacked polymorphic model for silicon content prediction in ironmaking process. IEEE Transactions on Instrumentation and Measurement, 70, pp.1-12.

[7] Tang, X., Zhuang, L. and Jiang, C., 2009. Prediction of silicon content in hot metal using support vector regression based on chaos particle swarm optimization. Expert Systems with Applications, 36(9), pp.11853-11857.

[8] Zhang, Q., Su, X., Ouyang, Z., Nie, W. and Liu, X., 2021, October. A model for predicting silica concentrate concertration based on Bayesian-LGC. In 2021 IEEE 9th International Conference on Smart City and Informatization (iSCI) (pp. 76-81). IEEE.

[9] Zhou, K., Hu, F. and Gong, J., 2021, May. Prediction of Silicon Content of Molten Iron in Blast Furnace Based on Particle Swarm-Random Forest. In 2021 33rd Chinese Control and Decision Conference (CCDC) (pp. 2814-2819). IEEE.

[10] Wang, W. and Ma, J., 2009, August. Principal component regression approach for forecasting silicon content in hot metal. In 2009 Fifth International Conference on Natural Computation (Vol. 2, pp. 590-593). IEEE.